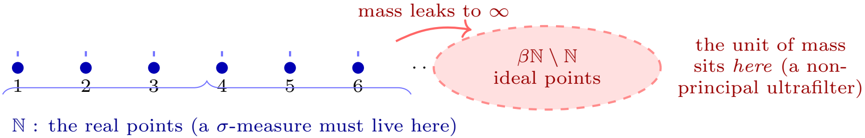
The Boolean leak (proved): “uniform over $\mathbb{N}$”

finite coherence everywhere, no σ-additive law



Every finite question is answered, consistently: $\mu(\{1,\ldots,k\}) = 0$ for each k, $\mu(\mathbb{N}) = 1$ — a perfectly coherent *finitely additive* charge (“pick a uniformly random natural number”). **Yet no σ-additive measure does this:** σ-additivity would force $1 = \mu(\mathbb{N}) = \sum_n \mu(\{n\}) = 0$. The mass is consistent on every finite set but sits, in the limit, on the *ideal* points $\beta\mathbb{N}\setminus\mathbb{N}$ — not on any real point. This is the CE phenomenon (Boolean, *proved*).

Why this is the stand-in. The σ-essential OML question asks for the *non-distributive* analogue of exactly this leak: a concrete σ-complete OML on which every *finite* subconfiguration is classically interpretable, yet the countable whole carries a state that no σ-additive mixture of dispersion-free states reproduces — the mass “leaking” off the genuine (σ-additive) valuations. **The open part**, and why it is hard: in the Boolean case the leak lives on $\beta\mathbb{N}\setminus\mathbb{N}$, a *distributive* (Boolean) boundary, and σ-additivity simply kills it by concentrating on points. The OML witness would need to host the same leak *off-centre*, on genuinely non-distributive structure where no such Boolean boundary absorbs it — which is the part nobody can presently build (or rule out).